

Naveen Anand Gunalan

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SUMMARY OF QUALIFICATIONS

- Hands-on AI platform engineer and software architect with 17+ years of engineering experience building production platforms, including AI systems, cross-platform developer SDKs, distributed systems, cloud services, and real-time embedded software.
- Shipped real-time interactive experiences across PC, console, mobile, and XR platforms.
- Strong expertise in XR, haptics, computer vision, and real-time human-computer interaction systems.
- Proven track record delivering AI-powered platforms from concept to production while defining technical strategy, architecture, and engineering roadmaps for multidisciplinary teams.
- Combines hands-on implementation with architecture leadership across AI, cloud, XR, embedded, and real-time systems.
- Proficient in Python, C, C++, C#, Objective-C, JavaScript/React, Kotlin, Java, SQL, and embedded systems.

CORE EXPERTISE

XR / Spatial Computing • Multimodal AI • Applied Generative AI • Game Development • Real-Time Human-Computer Interaction

AI / Generative AI: LLMs, RAG, embeddings, vector search, AI agents, multi-agent orchestration, tool calling, MCP, LLM evaluation, multimodal AI (text, vision, 3D geometry), generative modeling, agentic coding tools (Claude Code, Cursor, Copilot)

Backend, Cloud & Systems: REST/API design, event-driven architecture, PostgreSQL, vector databases; AWS, GCP, Azure, Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins, Bazel); low-latency networking (TCP/IP, UDP, WebSockets)

Embedded & Platforms: Embedded firmware, AOSP, OTA, power/thermal, hardware bring-up, DSP, MEMS & ultrasonic sensor arrays, computer vision, haptic/tactile rendering; SDKs, Unity, Unreal; Android, iOS, Windows, Linux, Meta Quest, Apple Vision Pro

Leadership: Architecture & technical strategy, engineering roadmaps, system design, engineering management, hiring & mentoring, OKRs, budget/P&L, cross-functional leadership, technical due diligence, executive communication

WORK EXPERIENCE

Architect / Senior Director of Engineering, Emerge

2018 to Present

- Built and led a 23-person global engineering organization spanning XR, embedded, DSP, hardware, and cloud; coordinated 30+ partners and delivered 20+ experiences supporting \$40M+ in funding.
- Owned Emerge's end-to-end platform strategy across Unity/XR, SDK networking, Android/AOSP, C/C++ beamforming, FPGA control, and GCP auth, delivering up to 8-player low-latency mid-air tactile experiences.
- Shipped Emerge Home, the flagship social product, across Meta Quest, iOS, and Android on custom ultrasonic hardware.
- Built an MCP-enabled RAG platform over 3,000+ engineering documents, combining document ingestion, semantic retrieval, vector search, tool calling, and multi-agent coding/debugging workflows, reducing engineering investigation, debugging, and documentation time by up to 3x.
- Built a multimodal Tactility Generation Model that turns text, screenshots, 3D geometry, material metadata, and user intent into tactile interaction specs: tactile points, edge-based sensations, texture feedback, and dynamic haptic effects.
- Wrote the C beamforming algorithm powering mid-air tactile sensation on Wave-1, Emerge's proprietary ultrasonic hardware.
- Cut release cycles 40% by automating AOSP and Unity app/SDK builds (Jenkins, Azure DevOps) across cloud and on-prem servers.
- Named inventor on 7+ granted patents.

Senior Software Engineer / Engineering Lead, Daqri

2015 to 2018

- Shipped 8+ AR applications for DAQRI devices, including Software Update, Turbine, Gallery, Onboarding, and WorkSense Tag for attaching contextual data and media to real-world assets.
- Architected Unity SDK extensions bridging managed C# and native layers, enabling API communication, hardware integration, and cross-platform deployment across Linux, Android, iOS, and Windows.
- Collaborated with OS, VIO, and SLAM teams on system integration, optimization, profiling, and debugging for AR hardware and software platforms.
- Led customer and developer engagements by educating partners on SDK capabilities, supporting integrations, and building demos for CES, AWE, Hannover, and client visits.

Senior Software Developer, Capture Interactive

2013 to 2015

- Shipped 10 AR/interactive apps and 5 games across Windows, Android, and iOS, spanning sports, casual, party, and trivia genres.

- Built VR and large-scale multi-screen interactive experiences, including an Oculus Rift DK2 Comic-Con demo and a 32-screen Crestron-based installation with MySQL/server-side integration.
- Software Engineer I, Cognizant Technology Solutions**

2009 to 2011
- Researched OPC UA protocol throughput and reliability as part of an R&D initiative.
 - Built a test-management web tool with client-server architecture using socket programming.
- Founder, Design For You**

2008 to 2009
- Created logos, animations, posters, business cards, and front- and back-end websites for a range of clients.

EDUCATION

- Full Sail University, B.S. Game Development** | GPA 3.56 | Winter Park, FL

Nov 2011 to Oct 2013
- PSG College of Technology, Anna University, B.E. Electronics & Communication Engineering** | GPA 3.20

Aug 2005 to May 2009

ACHIEVEMENTS

- Named inventor on 7+ granted patents.
- Received Course Director Award for 13 courses at Full Sail University.
- Awarded Advanced Achievement Award and named Valedictorian at Full Sail University.